



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LEARNER GUIDE

For

Certified Lean Six Sigma White Belt (CLSSWB)

Course Code: C142

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 7

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	1 July 2026	Initial release — CLSSWB Learner Guide for the one-day Lean Six Sigma awareness course.	Dr. Alfred Ang
2	23 September 2026	Guide rebuilt from the single-source content module and restructured to follow the DMAIC roadmap end to end, with exactly one hands-on activity per DMAIC phase (5 activities). All activities unified under one continuous scenario (the BrewBean Cafe morning rush) so each activity output feeds the next. Content simplified to White Belt awareness depth: sigma-level and DPMO calculation, MSA, FMEA, value stream mapping, Kano analysis, weighted solution-selection matrices and SPC control limits removed and replaced with recognition-level explanations; quick-reference section changed from formulas to the DMAIC roadmap; glossary aligned to the tools actually taught.	Dr. Alfred Ang
3	23 September 2026	Lab data pack added. Each of the five activities is now a self-contained folder carrying its mock dataset in CSV and Excel form, blank worksheet templates, a worked model answer and facilitator notes. The datasets are internally consistent across the five activities — the Activity 2 observation	Dr. Alfred Ang

		log is the evidence base for the Activity 3 Pareto, and the Activity 4 pilot results carry into the Activity 5 monitoring data — so the BrewBean Cafe story reconciles end to end. Lab steps rewritten to reference the specific data and template files the learner works from.	
4	23 September 2026	Labs restructured into the Tertiary Infotech house ACTIVITY format. labs/ is now activities/, with one folder per activity named 'NN - Title', each carrying the Facilitator Guide, Learner Worksheet and Checklist as DOCX and PDF alongside its data pack (CSV datasets, Excel workbook, blank templates and the model answer). Per-activity pointers updated to the new folder layout.	Dr. Alfred Ang
5	23 September 2026	Terminology aligned to the activity restructure: every learner-facing 'Lab N' reference now reads 'Activity N' across this guide and the slide deck, matching the activities/ folder names. Lesson Plan total-time footnote clarified to state its basis (480 minutes scheduled 9:30-18:30, excluding the 1-hour lunch break); the schedule rows themselves are unchanged.	Dr. Alfred Ang
6	23 September 2026	Completed the activity rename: the remaining learner-facing 'lab' wordings in the deck, this guide and the Lesson Plan now read 'activity'. Fixed the Lesson Plan's slide-	Dr. Alfred Ang

		reference lookup, which stopped matching when the deck's activity markers were renamed and had dropped the [slides N-M] references from the activity rows.	
7	23 September 2026	Activity step slides now carry the DMAIC phase in the eyebrow (ACTIVITY N - PHASE) instead of a character-truncated activity title.	Dr. Alfred Ang

TABLE OF CONTENTS

Introduction.....	6
Course Learning Outcomes.....	6
Before You Start.....	7
FOUNDATIONS — Lean Six Sigma Foundations (20%).....	8
DEFINE — Define — Understand the Problem (20%).....	8
Activity 1 — Define — Customer Requirements and the Problem Statement [Core].....	8
MEASURE — Measure — See What Is Really Happening (20%).....	10
Activity 2 — Measure — Map the Process and Spot the Waste [Core].....	10
ANALYZE — Analyze — Find the Cause (20%).....	11
Activity 3 — Analyze — Find the Root Cause [Core].....	12
IMPROVE — Improve — Fix the Cause (10%).....	13
Activity 4 — Improve — Choose and Pilot a Countermeasure [Core].....	14
CONTROL — Control — Hold the Gain (10%).....	15
Activity 5 — Control — Hold the Gain and Hand Over [Core].....	15
Quick Reference — The DMAIC Roadmap.....	17
Quick Reference — The Eight Wastes (DOWNTIME).....	18
Consolidating What You Learned.....	18
Glossary.....	19

Introduction

This Learner Guide accompanies the course Certified Lean Six Sigma White Belt (CLSSWB) (C142), conducted by Tertiary Infotech Academy Pte Ltd. This is a one-day Lean Six Sigma awareness course. It follows the DMAIC roadmap end to end — Define, Measure, Analyze, Improve, Control — with one hands-on activity in each of the five phases, and provides step-by-step instructions for every activity.

The course content is grounded in the body of knowledge published by The Council for Six Sigma Certification (CSSC) in 'Six Sigma: A Complete Step-by-Step Guide', so what you learn here matches the recognised White Belt standard.

As a White Belt you are being prepared to CONTRIBUTE to an improvement team — to understand the language, recognise waste, and support the tools your team uses. You are not expected to lead a project or perform the statistical analysis; those belong to the Yellow, Green and Black Belt levels.

Every activity uses one continuous scenario — the BrewBean Cafe morning rush, where customers queue up to 15 minutes and orders are sometimes made wrong. By the end of the day your five activity outputs form a complete improvement story: customer requirements and problem statement, process map and waste tally, root cause analysis, a selected countermeasure with a pilot plan, and a control plan.

Course Learning Outcomes

- LO1: Describe the core concepts of quality, Lean, Six Sigma and Lean Six Sigma, and the belt roles in an improvement team.
- LO2: Explain the five phases of the DMAIC roadmap and what happens in each.
- LO3: Identify customer requirements and a clear problem statement in the Define phase.
- LO4: Describe how a process is mapped and measured, and identify the eight wastes in the Measure phase.
- LO5: Identify likely causes of a problem using 5 Whys and Fishbone analysis in the Analyze phase.
- LO6: Describe common improvement and control actions that fix a cause and hold the gain.

Before You Start

What you need

- A laptop with a spreadsheet application (Microsoft Excel, Google Sheets or LibreOffice Calc), or simply pen and paper.
- A browser, for the interactive problem-solving tools used in the Analyze activity.

- The course slides and this Learner Guide, downloaded from <https://lms-tms.tertiaryinfotech.com>.
- A work process of your own to think about — the tools apply far better when the example is real.

The interactive problem-solving toolkit

Two browser-based tools are used during the activities. No installation or licence is required.

- 5 Whys — build and share a 5 Whys chain: <https://alfredang.github.io/5whys/>
- Fishbone Diagram — build an Ishikawa cause-and-effect diagram: <https://alfredang.github.io/fishbone/>

How the five activities fit together

- There are five activities — exactly one for each phase of DMAIC: Define, Measure, Analyze, Improve and Control.
- All five activities build on the same BrewBean Cafe scenario, so each activity's output becomes the next activity's input.
- Every activity is completed by everyone, and all five map directly to the DMAIC roadmap.

Conventions used in every activity

- Each activity states its objective, the deliverable you produce, the steps, and a check to confirm you are done.
- Tables shown in the steps can be built in a spreadsheet or on the worksheet provided.
- Where an activity uses an online tool, the tool URL is shown with the step.
- Keep every activity output — they combine into your final improvement package and are your revision material.

FOUNDATIONS — Lean Six Sigma Foundations (20%)

Quality · Lean · Six Sigma · Lean Six Sigma · Belt roles · The DMAIC roadmap

Key concepts

- What is Quality — Quality is meeting the customer's requirements — not just being free of defects.
- What is Lean — Lean removes waste so work flows faster to the customer.
- What is Six Sigma — Six Sigma reduces variation so results become consistent and predictable.
- Lean Six Sigma — The two combined: faster flow AND fewer defects, decided by data not opinion.

- Belt roles — White, Yellow, Green, Black and Master Black Belt — who does what on a team.
- The DMAIC roadmap — Define, Measure, Analyze, Improve, Control — the five-step improvement path.

DEFINE — Define — Understand the Problem (20%)

Voice of the Customer · CTQ · Problem statement · SMART goal · Project charter

Key concepts

- Voice of the Customer — Ask customers what they need, and record it in their own words.
- Critical to Quality — Turn each customer need into something specific you can measure.
- Problem statement — What is wrong, where, since when, and how big — never the solution.
- SMART goal — Specific, Measurable, Achievable, Relevant and Time-bound.
- Project charter — A one-page summary of the problem, goal, scope and team.
- Scope — Agreeing what is in and out keeps a small project small.

Activity 1 — Define — Customer Requirements and the Problem Statement

[Core]

Objective: Identify customer requirements and write a clear problem statement (A1, A2).

Goal: Start the BrewBean Cafe improvement the way every real project starts: find out what the customer actually wants, turn that into something measurable, and state the problem clearly enough that everyone agrees on it. The most common reason improvement projects fail is that the team never agreed what the problem was.

What you'll build

A VOC-to-CTQ table, a written problem statement, a SMART goal and a simple scope table. (Tools and techniques: Voice of the Customer, CTQ, problem statement, SMART goal, project scope.)

Your data pack

This activity ships with its own mock data for the BrewBean Cafe scenario. Everything below is in the activity folder, as CSV files and as one Excel workbook with a tab per sheet.

- DATA — Raw Voice of the Customer feedback (30 rows): Thirty pieces of unedited customer feedback collected by the BrewBean Cafe improvement team over two weeks — online reviews, comment cards and remarks made at the counter. This is

the raw material for Step 2 and Step 3: the learner reads it, extracts VOC statements and turns them into measurable CTQs.

- TEMPLATE — VOC-to-CTQ table (blank): Step 2 and Step 3. One row per VOC statement.
- TEMPLATE — Problem statement, SMART goal and scope (blank): Steps 4, 5 and 6.
- Excel workbook: A01-Data-Workbook.xlsx — every dataset and template above, one per tab.
- Model answer: model-answer.md — read it AFTER you attempt the activity.

Step-by-step

1. Read the BrewBean Cafe scenario, then open data/voc-raw-customer-feedback.csv (or the workbook) and read all 30 pieces of customer feedback. Underline every complaint you can find.
2. Using templates/voc-to-ctq-template.csv, write down five Voice of the Customer statements in the customer's OWN words, quoted from the Verbatim Comment column — do not paraphrase them yet.
3. Turn each VOC statement into a measurable CTQ requirement in the same template — every CTQ must carry a number AND a unit (e.g. 'served within 5 minutes').
4. In templates/problem-statement-and-scope-template.csv, write the problem statement: what is wrong, where, since when, and how big. Do NOT write a solution.
5. Check your problem statement against the SMART test and rewrite the goal so it is Specific, Measurable, Achievable, Relevant and Time-bound.
6. Fill the in-scope / out-of-scope rows at the bottom of the same template so the team knows where the project stops.

Check your work

Every CTQ has a number and a unit, your problem statement contains no solution, and your goal has a metric, a baseline, a target and a date.

Note: The Learner Worksheet, the Facilitator Guide, the Checklist, the data, the blank templates and the model answer for this activity are in the activities/01 - */ folder.

MEASURE — Measure — See What Is Really Happening (20%)

Process mapping · SIPOC · Types of data · Data collection · The eight wastes

Key concepts

- Process mapping — Draw the steps as they really happen, not as the manual says.
- SIPOC — A one-page overview: Suppliers, Inputs, Process, Outputs, Customers.
- Types of data — Discrete data you count; continuous data you measure on a scale.

- Data collection — Agree what to record, who records it, and when — before you start.
- Check sheets — A simple tally form is the easiest reliable way to collect data.
- The eight wastes — DOWNTIME — Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.

Activity 2 — Measure — Map the Process and Spot the Waste [Core]

Objective: Describe how the process runs and identify the eight wastes in it (A4, K2).

Goal: You cannot improve what you cannot see. Walk the BrewBean Cafe morning rush from the customer joining the queue to the drink being handed over, draw it as it really happens, and tally where time and effort are being lost. This is the phase that turns opinions about the problem into facts about the problem.

What you'll build

A SIPOC overview, a step-by-step process map with timings, and a completed waste tally sheet. (Tools and techniques: SIPOC, process mapping, types of data, check sheets, the eight wastes (DOWNTIME).)

Your data pack

This activity ships with its own mock data for the BrewBean Cafe scenario. Everything below is in the activity folder, as CSV files and as one Excel workbook with a tab per sheet.

- DATA — Morning rush observation log (60 rows): 60 customers observed end to end during one weekday morning rush at BrewBean Cafe. Wait time is measured from the customer joining the queue to the drink being handed over. This is THE baseline dataset for the course — Activity 3's Pareto is derived from the Delay Reason column and Activity 5's control chart compares back to it.
- DATA — Process step timings (15 rows): The average observed time for each step of the morning-rush process, with the team's value-add judgement left blank in the template. Use it for Step 3 and Step 4 of the activity.
- DATA — DOWNTIME waste tally (partially completed) (8 rows): The team's waste walk, half done. Three rows are filled in as worked examples; the learner completes the rest in Step 6 from the observation log and the process timings.
- TEMPLATE — SIPOC (blank): Step 2. Keep Process to 5-7 steps only.
- TEMPLATE — Process map with value-add judgement (blank): Steps 3 and 4.
- TEMPLATE — DOWNTIME tally sheet (blank): Step 6.
- Excel workbook: A02-Data-Workbook.xlsx — every dataset and template above, one per tab.
- Model answer: model-answer.md — read it AFTER you attempt the activity.

Step-by-step

7. Agree the start and stop points of the process — start: customer joins the queue; stop: customer receives the drink.
8. Complete templates/sipoc-template.csv: Suppliers, Inputs, Process (5-7 steps only), Outputs, Customers.
9. Open data/process-step-timings.csv. Copy the 15 steps into templates/process-map-template.csv with their observed times.
10. Mark each step VA or NVA in the template — the test is whether the customer would pay extra for THAT step — and write one line saying why.
11. Look at the columns in data/morning-rush-observation-log.csv and classify each: which figures are discrete (counted) and which are continuous (measured)?
12. Open data/downtime-waste-tally.csv — three rows are completed as worked examples. Using the observation log and the step timings as your evidence, complete the remaining five rows in templates/downtime-tally-template.csv.
13. Identify which single waste type you tallied most often and at which process step, then write down the baseline figures (average wait, longest wait, % over the 5-minute promise, drinks remade) — Activity 5 measures the improvement against them.

Check your work

Your SIPOC has all five columns filled, every process step has a time, and every waste on your tally sheet is tagged to one of the eight DOWNTIME types.

Note: The Learner Worksheet, the Facilitator Guide, the Checklist, the data, the blank templates and the model answer for this activity are in the activities/02 - */ folder.

ANALYZE — Analyze — Find the Cause (20%)

Root cause · 5 Whys · Fishbone diagram · Pareto chart · Variation

Key concepts

- Symptom vs cause — Treating the symptom makes the problem come back; treating the cause does not.
- Root cause — The cause that, once removed, stops the problem recurring.
- 5 Whys — Keep asking 'why?' until you reach something you can actually act on.
- Fishbone diagram — Sort possible causes into Manpower, Method, Machine, Material and Measurement.
- Pareto chart — Roughly 80% of the problem usually comes from 20% of the causes.
- Variation — Some variation is normal in the process; some has a specific assignable cause.

Activity 3 — Analyze — Find the Root Cause [Core]

Objective: Identify likely causes using 5 Whys and Fishbone analysis (A3, K2).

Goal: You now know WHERE the delay happens. This activity asks WHY it happens. Most teams jump straight from a symptom to a solution and end up fixing the wrong thing — the queue is long, so hire more staff — when the real cause was something else entirely. Here you drill past the symptom to a cause you can actually act on.

What you'll build

A completed 5 Whys chain, a Fishbone diagram with causes sorted by category, and a shortlist of likely root causes. (Tools and techniques: Symptom vs cause, 5 Whys, Fishbone (5M) diagram, Pareto chart, common vs special cause variation.)

Your data pack

This activity ships with its own mock data for the BrewBean Cafe scenario. Everything below is in the activity folder, as CSV files and as one Excel workbook with a tab per sheet.

- DATA — Delay reason Pareto data (6 rows): The Delay Reason column from the Activity 2 observation log, counted and ranked. This is the 'supplied Pareto chart' the activity's Step 5 refers to — the learner reads it rather than building it from scratch.
- DATA — Candidate causes and their evidence (12 rows): Every cause the team brainstormed, with whether any evidence actually supports it. Used in Step 7 to shortlist causes on evidence rather than opinion, and in Step 6 to judge common vs special cause.
- TEMPLATE — 5 Whys chain (blank): Steps 1 and 2. Stop when you reach something the team can act on.
- TEMPLATE — Fishbone (5M) diagram (blank): Steps 3 and 4. Problem goes in the head of the fish.
- TEMPLATE — Root cause shortlist (blank): Steps 6 and 7.
- Excel workbook: A03-Data-Workbook.xlsx — every dataset and template above, one per tab.
- Model answer: model-answer.md — read it AFTER you attempt the activity.

Step-by-step

14. Write the symptom from your Activity 2 baseline at the top of templates/5-whys-template.csv (e.g. 'customers wait an average of 8.9 minutes against a 5-minute promise').
15. Ask 'why does that happen?' down the template, each answer becoming the next question. Stop when you reach a PROCESS the team can act on — never at a person.
16. Open templates/fishbone-template.csv and write the problem in the head of the fish.
17. Brainstorm possible causes onto the five bones — Manpower, Method, Machine, Material, Measurement — then check them against data/cause-evidence-sheet.csv to see which ones the data actually supports.

18. Open data/delay-reason-pareto-data.csv — the delay reasons from the Activity 2 log, counted and ranked. Read the Cumulative % column and state which few causes account for most of the problem.
19. Using the last column of data/cause-evidence-sheet.csv, decide for each cause whether it is common cause (built into the process) or special cause (a one-off event).
20. In templates/root-cause-shortlist-template.csv, shortlist the two or three causes best supported by evidence — not by opinion.

Check your work

Your 5 Whys chain ends in something the team can actually act on, every Fishbone cause sits under one of the five categories, and each shortlisted cause is backed by an observation, not an opinion.

Note: The Learner Worksheet, the Facilitator Guide, the Checklist, the data, the blank templates and the model answer for this activity are in the activities/03 - */ folder.

IMPROVE — Improve — Fix the Cause (10%)

Generating solutions · 5S · Mistake proofing · Standard work · Piloting

Key concepts

- Generating solutions — Brainstorm against the proven cause — never against the symptom.
- Choosing a solution — Compare ideas on impact, effort and risk before committing.
- 5S — Sort, Set in order, Shine, Standardise, Sustain — a tidy workplace for physical or digital work.
- Mistake proofing — Poka-Yoke makes the error hard or impossible to make.
- Standard work — Write the better method down so everyone does it the same way.
- Piloting — Try the change small first, so mistakes stay cheap.

Activity 4 — Improve — Choose and Pilot a Countermeasure [Core]

Objective: Describe improvement actions that address the identified cause (A5, K2).

Goal: A cause is only worth finding if something changes because of it. Take the root causes you shortlisted in Activity 3, generate countermeasures against them, and pick one that is small enough to try next week. The discipline here is to solve the CAUSE you proved, not the symptom you started with.

What you'll build

A brainstormed countermeasure list, an impact/effort grid, one selected countermeasure and a simple pilot plan. (Tools and techniques: Brainstorming, impact/effort screening, 5S, mistake proofing (Poka-Yoke), standard work, piloting.)

Your data pack

This activity ships with its own mock data for the BrewBean Cafe scenario. Everything below is in the activity folder, as CSV files and as one Excel workbook with a tab per sheet.

- DATA — Countermeasure options with impact, effort and cost (12 rows): Twelve candidate countermeasures the team generated against the shortlisted root causes, scored for impact and effort. Used in Step 5 to plot the impact/effort grid and choose one.
- DATA — Pilot week results (6 rows): What actually happened when the chosen countermeasure was piloted for one week. The same 60-customer observation was repeated each day. Used in Step 7 to judge whether the pilot worked, and carried into Activity 5 as the new baseline.
- TEMPLATE — Countermeasure brainstorm (blank): Steps 1 and 2. At least six ideas, no judging yet.
- TEMPLATE — Impact/effort grid (blank): Step 5. Score each idea, then place it in a quadrant.
- TEMPLATE — Standard work and pilot plan (blank): Steps 6 and 7.
- Excel workbook: A04-Data-Workbook.xlsx — every dataset and template above, one per tab.
- Model answer: model-answer.md — read it AFTER you attempt the activity.

Step-by-step

21. Write your top root cause from Activity 3 at the top of templates/countermeasure-brainstorm-template.csv — every idea must address THIS cause.
22. Brainstorm at least six countermeasures without judging any of them yet, then compare your list with data/countermeasure-options.csv, which holds the twelve the team generated.
23. Check whether 5S would help: is anything at the counter hard to find, out of place or untidy?
24. Check whether mistake proofing would help: could the wrong order be made impossible rather than merely discouraged?
25. Using the Impact and Effort scores in data/countermeasure-options.csv, plot every idea on templates/impact-effort-grid-template.csv and pick ONE from the high-impact, low-effort corner.
26. In templates/standard-work-and-pilot-template.csv, write the improved method as standard work — short numbered steps with an unambiguous done/not-done check.
27. Complete the pilot plan rows of the same template: what changes, who runs it, what you will measure, the baseline to beat, and how you will know it worked. THEN open

data/pilot-week-results.csv and compare what actually happened — pay close attention to Thursday.

Check your work

Your selected countermeasure clearly addresses the root cause from Activity 3 (not the symptom), and your pilot plan names a measure, an owner and a review date.

Note: The Learner Worksheet, the Facilitator Guide, the Checklist, the data, the blank templates and the model answer for this activity are in the activities/04 - */ folder.

CONTROL — Control — Hold the Gain (10%)

Control plan · Visual management · SOPs · Team huddles · Handover

Key concepts

- Why Control matters — Without Control, processes quietly drift back to the old way.
- Control plan — What we measure, the target, how often, who owns it, and what to do if it slips.
- Visual management — Make performance visible so problems are noticed the same day.
- Standard operating procedures — Written instructions that lock in the improved method.
- Team huddles — Short regular stand-ups catch problems while they are still small.
- Handover — Give the improved process back to the people who run it every day.

Activity 5 — Control — Hold the Gain and Hand Over [Core]

Objective: Describe the control actions that sustain an improvement (A5, K2).

Goal: Most improvements work for a fortnight and then quietly disappear. Control is what stops that happening. Close the BrewBean Cafe story by deciding what will be measured from now on, who owns it, what happens when it slips, and how the improved process is handed back to the people who run it every day.

What you'll build

A completed control plan, a visual management board sketch, and a one-page summary of the whole DMAIC story. (Tools and techniques: Control plan, visual management, SOPs, team huddles, handover, one-page summary.)

Your data pack

This activity ships with its own mock data for the BrewBean Cafe scenario. Everything below is in the activity folder, as CSV files and as one Excel workbook with a tab per sheet.

- DATA — Post-improvement monitoring — four weeks (20 rows): Four weeks of daily average wait times after the pilot was made permanent. The learner uses this in Steps 1-3 to choose the control measure, set the target and write a reaction plan that would have caught the drift.
- DATA — Control plan — worked example (one row completed) (4 rows): The control plan format with a single worked row, so the learner can see the level of specificity expected before completing their own in Step 2.
- TEMPLATE — Control plan (blank): Steps 1, 2 and 3.
- TEMPLATE — SOP and daily huddle plan (blank): Steps 5 and 6.
- TEMPLATE — One-page DMAIC summary (blank): Step 7. The whole project on one page.
- Excel workbook: A05-Data-Workbook.xlsx — every dataset and template above, one per tab.
- Model answer: model-answer.md — read it AFTER you attempt the activity.

Step-by-step

28. Open data/post-improvement-monitoring.csv — four weeks of daily figures after the pilot was made permanent. Choose the ONE measure that tells you the improvement is still working.
29. Look at data/control-plan-example.csv for the worked first row, then complete templates/control-plan-template.csv: measure, target, how often it is checked, where it is recorded and who owns it.
30. Write the reaction plan — the exact steps to take when the measure misses target. Then find week 3 in the monitoring data and state which day your plan would have triggered on.
31. Sketch a simple visual board that would have made the week-3 drift visible to the whole shift on day one.
32. In templates/sop-and-huddle-template.csv, turn your Activity 4 standard work into a short SOP the cafe could actually pin up.
33. Complete the huddle rows of the same template: who attends, what is reviewed, who runs it and how long it lasts.
34. Using templates/one-page-summary-template.csv, summarise the whole project — problem, cause, countermeasure, result and control — and name who you hand it over to and when.

Check your work

Your control plan names a measure, a target, a frequency and an owner; your reaction plan says what to DO when it slips; and your one-page summary tells the complete DMAIC story from problem to handover.

Note: The Learner Worksheet, the Facilitator Guide, the Checklist, the data, the blank templates and the model answer for this activity are in the activities/05 - */ folder.

Quick Reference — The DMAIC Roadmap

What each phase asks, and what it delivers

- Define — 'What problem are we solving, and for whom?' Delivers the Voice of the Customer, the CTQ requirements, a problem statement, a SMART goal and an agreed scope.
- Measure — 'How big is the problem, really?' Delivers a SIPOC, a process map, a data collection plan and a baseline of how the process performs today.
- Analyze — 'Why is this happening?' Delivers the shortlisted root causes, supported by 5 Whys, a Fishbone diagram and the data collected in Measure.
- Improve — 'What change will actually fix it?' Delivers a selected countermeasure, standard work for the new method, and a pilot plan.
- Control — 'How do we make the gain stick?' Delivers a control plan, visual management, an SOP and a handover to the process owner.

A note on Six Sigma measurement

You will hear practitioners refer to a process's 'sigma level' and to DPMO (Defects Per Million Opportunities) — the measure Six Sigma uses to describe how often a process produces a defect. The well-known Six Sigma target is 3.4 defects per million opportunities, which is why the method carries that name.

As a White Belt you are expected to RECOGNISE these terms and understand what they describe. Calculating DPMO and converting it to a sigma level is done at Yellow Belt and above, and is not assessed on this course.

Quick Reference — The Eight Wastes (DOWNTIME)

- D — Defects: output that fails the requirement and must be corrected or redone.
- O — Overproduction: producing more, or earlier, than the customer needs.
- W — Waiting: work or people idle, waiting for the next step, an approval or information.
- N — Non-utilised talent: skills and ideas of people not being used.
- T — Transport: unnecessary movement of materials, work items or information between places.
- I — Inventory: work in progress, backlogs and queues sitting between steps.
- M — Motion: unnecessary movement of people, or switching between systems and screens.

- **E** — Extra-processing: doing more work to the output than the customer requires or values.

Consolidating What You Learned

- Keep these slides, this Learner Guide and your activity outputs — they are your reference after the course.
- Re-read your own activity outputs; they follow one continuous scenario from Define through to Control.
- Be ready to define quality, Lean, Six Sigma and Lean Six Sigma, and explain how they differ.
- Be ready to describe the belt roles and say where a White Belt contributes to an improvement team.
- Be ready to name the eight wastes (DOWNTIME) and give a workplace example of each.
- Be ready to explain each DMAIC phase, what it delivers and which tools belong to it.
- Be ready to translate a VOC statement into a measurable CTQ requirement.
- Be ready to write a problem statement that describes the problem without naming a solution.
- Be ready to explain how the Fishbone diagram and 5 Whys are used together to find a root cause.
- Be ready to describe what a control plan must contain to hold an improvement in place.
- Re-work the activities from memory — being able to produce the tools unaided is the best preparation.

Glossary

- **Lean** — A method to maximise customer value by systematically identifying and removing waste.
- **Six Sigma** — A data-driven method to reduce variation and defects; the target is 3.4 defects per million opportunities.
- **Lean Six Sigma** — The combined method — Lean improves speed and flow, Six Sigma improves consistency and accuracy.
- **DMAIC** — Define, Measure, Analyze, Improve, Control — the Six Sigma improvement roadmap.
- **PDCA** — Plan, Do, Check, Act — a lighter improvement cycle used for small, fast improvements.
- **VOC** — Voice of the Customer — customer needs and expectations expressed in the customer's own words.
- **CTQ** — Critical to Quality — a specific, measurable requirement translated from a VOC statement.

- **SIPOC** — Suppliers, Inputs, Process, Outputs, Customers — the macro 'as-is' process map.
- **Muda** — The Japanese term for waste — any activity that consumes resource but creates no customer value.
- **DOWNTIME** — Mnemonic for the eight wastes: Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.
- **Value-added (VA)** — An activity the customer would be willing to pay for because it changes the product or service.
- **Non-value-added (NVA)** — An activity that consumes resource but adds nothing the customer values — pure waste.
- **Defect** — An output that fails to meet the CTQ requirement.
- **Common cause variation** — Natural, always-present variation built into the process; addressed by changing the process.
- **Special cause variation** — Unusual variation traceable to a specific assignable event; addressed by investigating that event.
- **Pareto principle** — The 80/20 rule — roughly 80% of effects come from 20% of causes.
- **5 Whys** — Repeatedly asking 'why' to drill from a symptom down to an actionable root cause.
- **Fishbone (Ishikawa) diagram** — A cause-and-effect diagram organising candidate causes into categories such as the 5Ms.
- **5M categories** — Manpower, Method, Machine, Material, Measurement — standard Fishbone categories.
- **DPMO** — Defects Per Million Opportunities — the defect-rate measure Six Sigma uses. Recognise the term; calculating it is a Yellow Belt skill.
- **Yield** — The percentage of output produced without defects.
- **Sigma level** — A yardstick for process performance derived from DPMO; the Six Sigma target is 3.4 DPMO.
- **Discrete data** — Data you obtain by counting, such as the number of orders made wrong.
- **Continuous data** — Data you obtain by measuring on a scale, such as waiting time in minutes.
- **Check sheet** — A simple structured tally form used to collect process data consistently.
- **5S** — Sort, Set in order, Shine, Standardise, Sustain — a Lean method for organising the workplace.
- **Poka-Yoke** — Mistake proofing — designing the process so an error is difficult or impossible to make.
- **Standard work** — The documented current best-known method for performing a task.
- **Pilot** — A small-scale trial of a change, run before full rollout so mistakes stay cheap.

- **Control plan** — The document naming the metric, target, monitoring method, frequency, owner and reaction plan.
- **Reaction plan** — The agreed steps to take when a controlled measure misses its target.
- **Visual management** — Making process performance visible at a glance, so problems are noticed the same day.
- **SOP** — Standard Operating Procedure — written instructions that lock in the improved method.